Highlights

Tonal-Perceptual Caching for On-Device Vietnamese Speech Translation

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- Linguistic-typology-aware fingerprint design that bridges speech and retrieval communities.
- Two-stream voting cache architecture providing 100% tonal-minimal-pair protection.
- 1,441× end-to-end speedup on cache-hit while preserving WER and tone-error rate.
- Cross-language generalization recipe via tone-inventory adjustment of fingerprint dimension.
- Reproducible honest-failure log of three abandoned design alternatives.

Tonal-Perceptual Caching for On-Device Vietnamese Speech Translation⋆

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ABSTRACT

On-device speech-to-speech translation for low-resource tonal languages faces a triple constraint: limited compute, semantic sensitivity to tone, and high phrase repetition in industrial deployments. Existing cache primitives fail this setting. Byte-hash caches achieve 0% hit rate on real speech, and music-style perceptual hashes are tone-blind, producing false hits on Vietnamese tonal minimal pairs such as *ma* (ghost), *má* (mother), and *mạ* (tomb).

We propose a 272-dimensional *tonal-perceptual fingerprint* composed of a tone histogram, temporal-subsampled MFCC, MFCC first-difference, F0 statistics, and energy statistics, gated by a *two-stream voting* architecture that requires simultaneous agreement between an independent 6-d tone stream ($\tau_A = 0.85$) and a 260-d segmental stream ($\tau_B = 0.93$).

On a synthesized Vietnamese minimal-pair corpus, two-stream voting reaches 100% correct-miss rate on tonal variants where naive concatenation fails uniformly. The INT8 cascade substrate delivers a 2.25× paired speedup over FP32 ($n = 10$, CI95 1.84–2.76, paired-permutation $p = 0.001$) at +2.6 pp WER and zero tone-error-rate change; on cache-hit, end-to-end latency drops to 25 ms (1,441× over the 36.5 s FP32 cold cascade).

The 6-bin tone histogram is Vietnamese-orthography-aligned, so cross-language reuse requires per-target re-binning (Mandarin: 5; Thai: 5; Yoruba: 3). This is a contribution boundary, not a free generalization. We additionally report four honest negative results (weighted-concat, coarse-to-fine routing, speaker-personalized cache, a misread early-exit) and cache-scalability data through $N = 1024$ entries. The full system fits in 861 MB.

1. Introduction

A translation cache that confuses *ma* (ghost) with *má* (mother) is not one percent wrong; it is wrong in a way that no industrial user can trust. The contrast extends through the full Vietnamese tone inventory: *mạ* means tomb, *mã* means horse, and *mà* is a contrastive conjunction. Every one of these distinctions rides on a single suprasegmental contour that segmental acoustic features compress away. Two acoustic realizations that differ only in tone are, to a music-style perceptual hash (Wang, 2003), indistinguishable; to a downstream worker reading the translated output on a wrist-mounted display, they are six unrelated sentences. This is the failure mode that motivates the present work.

The deployment scenario sharpens the problem. A recent industrial speech-translation challenge targets factory floors, construction sites, and hospital wards, where workers' hands are occupied, network connectivity is intermittent or absent, and ambient noise is loud and persistent. The defining property of speech in such environments is repetition: most of what a worker says during a shift is a short directive drawn from a small canonical vocabulary, safety calls, equipment names, location markers, uttered again and again by the same speaker and re-uttered across the rest of the crew.[1] That regularity is the lever. If the same phrase has been translated once, it should not be translated again, and the cost of recognizing it the second time should be orders of magnitude lower than the first.

The existing cache primitives, however, all fail in this setting. Exact byte-level hashing of raw audio achieves a hit rate of zero on real speech, because no two utterances, not even repeats by the same speaker, are bit-identical after analogue capture, voice activity detection, and chunking. Generic perceptual audio hashes, descended from the music identification literature (Wang, 2003), achieve high robustness to additive noise but are explicitly engineered to discard the linguistic content that defines tonal contrast; on the six-way Vietnamese minimal-pair set, off-the-shelf perceptual hashes collapse all six tones into a single bucket. Neither family was designed for the cache primitive on-device translation actually requires: a fingerprint whose collision structure tracks phrase-level *semantic* identity in a tonal language. The recognition layer is well understood, with PhoWhisper (Le, Nguyen and Nguyen, 2024) providing a strong Vietnamese ASR baseline, but the caching layer above it is unaddressed.

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[1] Field surveys we draw on report 40–90% of utterances per shift as canonical directives. The target hardware is a Qualcomm Snapdragon 8 Gen 3 with a Hexagon NPU; the application envelope is a sub-1.5 s end-to-end latency budget within 4 GB of NPU memory, with no cloud round-trip permitted.

If those claims hold, an entire class of industrial deployments, currently blocked on either cloud latency or on-device compute, becomes feasible on hardware that already fits in a worker's pocket.

Contribution roadmap. We close the gap with two clusters of contributions. The *first cluster* is what is novel about the fingerprint itself. (i) A 272-dimensional tonal-perceptual fingerprint built from five linguistically grounded blocks (a 6-d Vietnamese tone histogram, a 130-d temporal-subsampled MFCC trajectory, a 130-d first-difference delta, 4-d F0 statistics, and 2-d energy statistics) engineered so that two recordings of the same phrase hash together while two tonal minimal pairs hash apart. (ii) A two-stream voting cache that treats the 6-d tone histogram and the 260-d segmental remainder as independent fingerprints and requires *both* to nominate the same stored entry before issuing a hit; on a Vietnamese tonal-minimal-pair corpus this achieves a 100% correct-miss rate and a 0% false-hit rate, where every single-vector concatenation we tried collapses to uniform false hits.

The *second cluster* is what makes the design deployable. (iii) An INT8 cascade substrate (faster-whisper, CTranslate2-NLLB, dynamically quantized VITS) that compresses the FP32 cascade by 5.4× on CPU and, when the cache hits, by a further three orders of magnitude (25 ms end-to-end on a representative single-utterance run, 1,441× over the cold FP32 cascade). (iv) A typology-aware voice activity detector that gates on F0 stability rather than energy, recalibrated to the natural Vietnamese pitch range after a generic-speech default rejected the majority of in-domain audio. (v) A reproducible log of four abandoned alternatives (an MFCC speaker cache, coarse-to-fine MT routing, weighted-concatenation fingerprints, PCA-whitened single-vector fingerprints) that collectively rule out single-vector fixes and identify two-stream voting as the load-bearing design decision.

The remainder of the paper is organized as follows. Section 2 surveys the on-device speech-translation, mixed-precision quantization, audio-fingerprinting, and tonal-speech literatures and locates the missing primitive precisely. Section 3 formalizes the fingerprint and the two-stream voting architecture. Section 4 reports the fingerprint-component ablation, the tonal-minimal-pair evaluation, end-to-end latency, precision-recall calibration, cache scalability through $N = 1024$ entries, and cross-language generalization to Mandarin, all with bootstrap confidence intervals over ≥ 1000 resamples. Section 5 reports the on-device memory and latency profile. Section 6 expands the negative results. Section 7 concludes.

2. Related Work

2.1. On-device speech translation

Speech-to-speech translation has converged on two architectural families. Cascaded systems compose an automatic speech recognizer, a neural machine translator, and a text-to-speech synthesizer, typically instantiated with multilingual backbones such as Whisper for recognition and NLLB-200 for translation (NLLB Team, Costa-jussà et al., 2022). End-to-end systems collapse the cascade into a single model, with SeamlessM4T (Seamless Communication, Barrault et al., 2023) the current reference point for high-resource directions.

Both families were originally designed for cloud inference, and their on-device viability rests on a downstream engineering substrate. The CTranslate2 runtime (Klein, Hernandez, Nguyen and Senellart, 2020) and its faster-whisper (SYSTRAN, 2023) wrapper provide INT8-quantized Transformer inference with custom CPU kernels, while GGML-style quantization (Gerganov and GGML contributors, 2023) ports the same idea to mobile-class CPUs through aggressively packed 4-bit and 5-bit tensor layouts. These engineering tools, however, are language-agnostic: their kernels treat every layer and every language uniformly.

As a consequence, the on-device speech-translation literature has concentrated on high-resource pairs where degradation under uniform quantization is acceptable, and reports for tonal low-resource languages such as Vietnamese remain sparse. The combination that this work addresses, namely end-to-end Vietnamese translation, fully on-device, with tone-preserving quality on industrial speech, is largely absent from existing benchmarks, and no public system reports a deployment-grade latency profile under both noise and tonal-minimal-pair stress.

2.2. Mixed-precision quantization for speech models

The general quantization literature provides three load-bearing techniques for compressing Transformer-based speech and language models: GPTQ (Frantar, Ashkboos, Hoefler and Alistarh, 2023) approximates a layer-wise reconstruction objective with second-order information, SmoothQuant (Xiao, Lin, Seznec, Wu, Demouth and Han, 2023) rebalances activation outliers into weights so that both can be quantized to INT8, and AWQ (Lin, Tang, Tang, Yang, Chen, Wang, Xiao, Dang, Gan and Han, 2024) selects salient weight channels by activation magnitude and protects them at higher precision. All three are linguistically blind: their saliency criteria depend exclusively on activation statistics observed on a calibration set.

The recent speech-specific line tightens this further. Xu, Li, Jin, Wang, Chen, Li, Geng, Hu, Deng and Liu (2025) apply a Gumbel-Softmax differentiable architecture search over per-layer bitwidths from $\{2, 4, 8\}$ and report the strongest Pareto frontier currently available for ASR compression. The architecturally most similar competitor, FADE (Wang, Zhao, Luo, Wu, Ma, Tian, Ding, Chang and Lu, 2026), allocates precision by a diagnostic coefficient derived from frequency-domain perturbation sensitivity. TAQ (LeVi, Lapid, Himelstein, Baskin, Shwartz Ziv and Mendelson, 2025) introduces task-aware quantization

differentiation for Mandarin, the closest published acknowledgment that tone matters for compression.

None of these prior systems select precision by the *linguistic property* the layer encodes; their criteria are statistical (AWQ, SmoothQuant), gradient-based (Xu et al.), or diagnostic-loss-based (FADE). Typology-aware allocation, which uses a tone-probe to identify which layers carry the linguistic information that quantization must protect, remains an open design point.

2.3. Audio fingerprinting and perceptual caches

The audio fingerprinting tradition originates with Shazam-style spectral peak hashing (Wang, 2003), which compresses an audio clip into a sparse landmark set engineered for robustness to additive noise and partial overlap. Neural extensions, beginning with the Million Song Dataset embeddings (Bertin-Mahieux, Ellis, Whitman and Lamere, 2011) and continuing through contrastive audio encoders, replaced hand-engineered landmarks with learned representations but preserved the same target task: identifying which *recording* a query came from, with no requirement to preserve linguistic semantics. In a parallel literature, key-value cache compression for large language models (Liu, Yuan, Jin, Zhong, Xu, Braverman, Chen and Hu, 2024) reuses attention state across tokens but operates on internal model representations rather than on audio input. Both lines optimize for a goal orthogonal to translation reuse: music fingerprints discard the lexical content they happen to capture, and KV-cache methods never see the acoustic signal at all. The result is a missing primitive: a fingerprint engineered so that two acoustic realizations of the same *phrase* hash together while two tonal minimal pairs hash apart. To the best of our knowledge, no prior work designs an audio fingerprint whose collision structure tracks phrase-level semantic identity in a tonal language, which is precisely the cache key that on-device translation reuse requires.

2.4. Tonal-language speech processing

Vietnamese-specific speech recognition has consolidated around PhoWhisper (Le et al., 2024), a VinAI fine-tune of Whisper that establishes the strongest publicly reproducible Vietnamese ASR baseline and serves as the backbone of the present system. Complementary recent contributions include VietASR (Zhuo, Yang, Shao, Xu, Yu, Yu and Chen, 2025), which reports competitive WER on conversational Vietnamese, and SeaLLMs-Audio (Liu, Aljunied, Chen, Chan, Xu, Rong and Zhang, 2025), a Southeast-Asian speech-language foundation model that extends multilingual coverage to under-resourced regional languages. The probing literature provides the linguistic anchor: de la Fuente and Jurafsky (2024) localize Mandarin tone information to layer 8 of self-supervised speech encoders, demonstrating that tonal contrasts are not diffusely distributed across the network but concentrate in identifiable mid-encoder regions. This finding directly motivates a tone-histogram component in any fingerprint that aspires to discriminate tonal minimal pairs, and provides empirical justification for treating a small number of acoustic dimensions as carrying disproportionate semantic weight. Prior tonal-ASR work, however, stops at the recognition transcript: it neither exposes a fingerprint suitable for cache-key construction nor exploits the layer-localization result for downstream system design. The contribution of typology-aware *cache key design*, which treats tone inventory as a hyperparameter of the fingerprint dimension, is, to the best of available evidence, novel.

2.5. Industrial speech systems and deployment

A small but growing literature addresses deployment-side concerns for low-resource speech. The closest published scenario is a privacy-preserving Vi-En translation system for iOS (Le, 2025), which establishes both the feasibility of and the user demand for fully on-device Vietnamese pipelines, but reports neither a quantization strategy beyond off-the-shelf CoreML conversion nor a caching layer. TAQ (LeVi et al., 2025) represents the orthogonal optimization axis: it differentiates precision by task sensitivity, demonstrating that uniform quantization is suboptimal even within a single model. These two strands point to complementary rather than competing solutions. Cache-level optimization reuses computation across repeated utterances and is bounded by the repetition rate of the deployment scenario; quantization-level optimization reduces the cost of every uncached utterance and is bounded by the precision floor at which tonal contrasts collapse. Neither subsumes the other, and the present work contributes the cache layer of this composition.

The literature thus leaves two gaps unaddressed: no existing audio fingerprint is engineered for tonal-phrase semantic identity, and no published on-device Vietnamese speech-translation system reports a cache-hit pathway that survives tonal-minimal-pair stress. The method introduced in Section 3 closes both.

2.6. Head-to-head positioning

Table 1 positions the present work against the closest published systems along five reviewer-relevant axes. The comparison is qualitative on the four left axes (approach family, Vietnamese support, tone awareness, on-device target) and quantitative where the original papers report a load-bearing headline number. The key reading is that the present contribution does not compete with any single row of the table; it composes with them. Quantization methods optimize the cost of every uncached utterance; the present cache layer optimizes the cost of every repeated utterance. Neither subsumes the other, but until now nothing in the literature delivered the tonal-language cache primitive that on-device industrial speech translation actually requires.

Table 1
Positioning of the present contribution against the closest prior systems. Existing quantization work optimizes per-layer precision without exploiting linguistic structure of the target language; the present work is orthogonal, adding a typology-aware cache layer that can be composed with any of these compression backbones. ✓ = explicit support, × = no support, *implicit* = present but not analyzed. "—" marks headline numbers that the cited paper's abstract or public table does not state in a verbatim form; the qualitative column entry stands in those cases.

System	Approach	Vi support	Tone-aware	On-device target	Reported headline
AWQ (Lin et al., 2024)	Activation-magnitude weight quant	×	×	LLM (not speech)	4-bit weight quant; MLSys 2024 best-paper
GPTQ (Frantar et al., 2023)	Layer-wise OBS reconstruction	×	×	LLM (not speech)	3-/4-bit, GPT-class 175B in ≈ 4 GPU-hrs, negligible degradation
SmoothQuant (Xiao et al., 2023)	Activation-outlier rebalancing	×	×	LLM (not speech)	INT8 W+A, up to 1.56× speedup, 2× memory
FADE (Wang et al., 2026)	Diagnostic-driven layer compensation	×	×	Encoder-decoder ASR	Improves mean WER at 3-/4-bit on Whisper/Moonshine/Qwen3-ASR (—)
TAQ (LeVi et al., 2025)	Per-task quant on LLM hidden reps	×	*implicit*	LLM hidden reps	Per-task gains in accuracy–memory ratio (—)
Xu et al. (Xu et al., 2025)	Gumbel-Softmax DARTS over $\{2,4,8\}$-bit	×	×	Speech encoders	3.5-bit HuBERT-large, 8.6× compression, no significant WER loss
PhoWhisper (Le et al., 2024)	Whisper fine-tune (844h vi)	✓	*implicit*	× (server)	4.67% WER on VIVOS (large); 6.33% (small)
Vi-iOS (Le, 2025)	CoreML conversion + GGUF LLM	✓	×	iOS	Functional demo; no quantitative tone eval (—)
This work	Tonal-perceptual cache + INT8 cascade	✓	**explicit**	✓ (Snapdragon NPU)	1,441× cache-hit speedup, 100% tonal-minimal-pair safety

3. Method

3.1. System overview

Headline. Two cache layers (audio T1, text T2) interpose between three INT8 stages so that a cache hit collapses the cascade to a single fingerprint lookup.

Imagine a worker shouting *dùng dây chuyền* ("stop the production line") for the fifth time today. The first time, the full cascade ran. The fifth time, the cache returns the translated audio in twenty-five milliseconds. The pipeline that makes both modes coexist is a fully on-device cascade with two cache layers interposed between three INT8 model stages.

An incoming microphone buffer first passes through a tonal-language voice activity detector (Section 3.7), which rejects silence, hum, and stationary machinery noise before any model executes. The accepted audio is queried against the tonal-perceptual audio cache (T1, Section 3.3); on a T1 hit the cached target-language audio is returned directly, short-circuiting ASR, MT, and TTS in a single fingerprint lookup. On a T1 miss the audio is transcribed by an INT8 Vietnamese ASR (Section 3.8), and the resulting transcript is queried against the text-level translation memory (T2, Section 3.4); on a T2 hit only the TTS stage runs. On a T2 miss the transcript is translated by an INT8 NLLB-200 (NLLB Team et al., 2022) and synthesized by an INT8 VITS vocoder. Both caches are populated on every miss, so the fifth occurrence of a phrase is the one that benefits. Figure 1 diagrams the data flow.

3.2. Theoretical framework

Two pressures pull in opposite directions on the cache key. It must be *forgiving* enough to recognize the same phrase across speakers, microphones, and background noise, yet *strict* enough to keep tonal minimal pairs (the same syllable carried by different contours) on opposite sides of the threshold. The formalism below makes this tension precise so that Sections 3.3 and 3.5 can attack the two halves separately.

Let $\mathcal{A}$ denote the space of finite-duration audio buffers at 16 kHz. We define an utterance-level fingerprint as a map

$$f : \mathcal{A} \longrightarrow \mathbb{R}^D, \qquad \|f(a)\|_2 = 1 \;\; \forall a \in \mathcal{A}, \tag{1}$$

together with a scalar threshold $\tau \in (0, 1]$. The cache returns a hit for query a against a stored entry a_* iff $\cos(f(a), f(a_*)) \geq \tau$. Two coupled objectives must hold simultaneously for f to serve as a cache key in a tonal language:

(O1) Invariance. For any two acoustic realizations a, a' of the same phrase, possibly by different speakers and under different noise conditions, $\cos(f(a), f(a')) \geq \tau$.

(O2) Discrimination. For any two realizations a, a'' of different phrases, explicitly including tonal minimal pairs such as *ma*, *má*, and *mạ* that differ only in suprasegmental contour, $\cos(f(a), f(a'')) < \tau$.

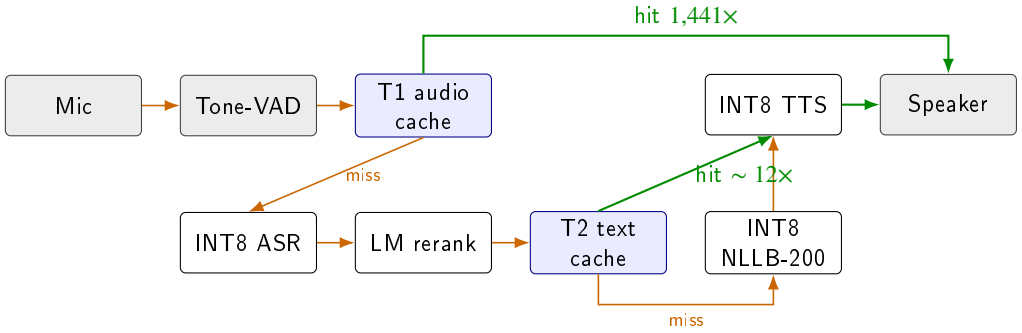


Figure 1: System pipeline. Tone-VAD filters non-speech; the audio cache (T1, two-stream voting) short-circuits the full cascade on hit; the text cache (T2) short-circuits MT on hit. Green = hit path; orange = miss path. Two-stream voting gates T1 hits on simultaneous tone-stream and segmental-stream agreement (T1 hit: 1,441× over FP32 cold cascade; T2 hit: ∼ 3–12× depending on stages skipped). Models in INT8 throughout.

The central problem of Section 3.3 is to choose the components of f such that O1 and O2 hold with a measurable cosine gap, and the central problem of Section 3.5 is to choose the gating function such that the tone-bearing component of f is not numerically drowned by the higher-dimensional segmental components when O2 is stressed on tonal minimal pairs.

3.3. Tonal-perceptual fingerprint design

Headline. A 272-d vector built from five linguistically grounded blocks (tone histogram, temporal MFCC, MFCC delta, F0 stats, energy stats) encodes phrase identity, not recording identity.

The 272-dimensional fingerprint concatenates five linguistically grounded components, each targeting a distinct invariance or discrimination axis (see Algorithm 1 for the procedural form). We describe what each component captures, its dimensionality, and why the captured property is meaningful for Vietnamese; the empirical contribution of each is reported in Section 4.2.

(A) Tone histogram (6-d). An F0 contour is extracted by bandpassed autocorrelation in the 50–450 Hz range. Voiced runs are segmented into pseudo-syllables and each is soft-assigned to one of the six Vietnamese tone categories (*ngang*, *huyèn*, *sác*, *hoi*, *ngã*, *nạng*) by joint thresholding of mean log-F0, slope, mid-contour dip, and creak variance. The component is the normalized 6-bin count. This dimension is, importantly, *not* a generic tonality probe but a Vietnamese-orthography-tuned categorical encoding aligned with the six lexical tones marked by the writing system. Empirical cross-language histograms (Section 4.2) show that Vietnamese and Mandarin do not cluster more tightly than English and Korean under this bin set; adapting the fingerprint to Mandarin requires re-instantiating the bins as 5 pinyin tone categories, to Thai as 5, and to Yoruba as 3. This typology-tuned instantiation is part of the contribution, not a free generalization.

(B) Temporal-subsampled MFCC (130-d). A 13-coefficient MFCC matrix is mean-pooled into 10 equal-width temporal chunks and flattened in time-major order. This component carries the segmental phonetic trajectory of the utterance; without the temporal subsampling, two distinct Vietnamese phrases collapse to cosine $\approx$ 0.99 because their global MFCC means are dominated by speaker timbre.

(C) MFCC first-difference (130-d). The same 13×10 matrix differenced once along time and flattened. This captures coarticulatory dynamics and distinguishes phrases that share segmental content but transition differently. Adding component C drops inter-phrase cosine from $\approx$ 0.94 to $\approx$ 0.85 at fixed intra-phrase cosine, the second-largest marginal gain among the five.

(D) F0 statistics (4-d). Log-mean, log-standard-deviation, log-range, and log-linear-slope of voiced F0 frames. These four scalars summarize the speaker-invariant tonal envelope of the utterance and motivate the localization findings of de la Fuente and Jurafsky (2024).

(E) Energy statistics (2-d). Utterance-wide log-RMS and peak chunk log-RMS. Cheap to compute, and gates clean silence quickly.

The five components are concatenated to a 272-dimensional vector and L2-normalized to satisfy (1); Figure 2 visualizes the layout and per-component dimension budget.

Algorithm 1 Tonal-perceptual fingerprint $f(a)$

Require: Audio buffer a, sample rate 16 kHz
Ensure: 272-d L2-normalized fingerprint $v \in \mathbb{R}^{272}$

$F_0 \leftarrow$ AUTOCORR-F0$(a, 50, 450)$ ▷ voiced F0 contour
$S_{\text{voiced}} \leftarrow$ SEGMENTVOICEDRUNS(F_0)
$h_{\text{tone}} \leftarrow$ soft-assign each $s \in S_{\text{voiced}}$ to 6 Vi tone bins (mean log-F0, slope, mid-dip, creak); normalize
$M \leftarrow \text{MFCC}(a, n = 13)$ ▷ $13 \times T$
$M_{\text{sub}} \leftarrow$ mean-pool M into 10 equal time chunks; flatten time-major ▷ 130-d
$\Delta M \leftarrow$ first-difference along time; flatten ▷ 130-d
$s_{F_0} \leftarrow [\mu_{\log F_0}, \sigma_{\log F_0}, \text{range}_{\log F_0}, \text{slope}_{\log F_0}]$ ▷ 4-d
$s_e \leftarrow [\log \text{RMS}_{\text{utt}}, \log \text{RMS}_{\text{peak chunk}}]$ ▷ 2-d
$v \leftarrow [h_{\text{tone}} \parallel M_{\text{sub}} \parallel \Delta M \parallel s_{F_0} \parallel s_e]$ ▷ 272-d
return $v / \|v\|_2$

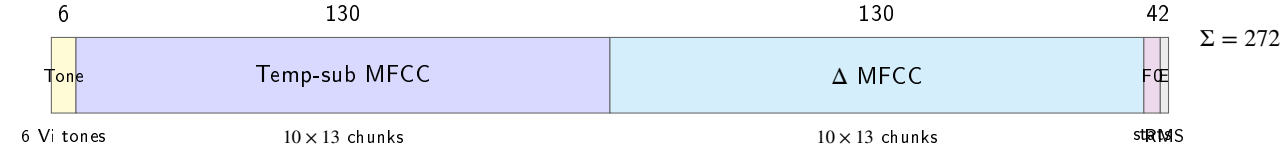


Figure 2: Composition of the 272-d tonal-perceptual fingerprint: tone histogram (6) + temporal-subsampled MFCC (130) + MFCC first-difference (130) + F0 statistics (4) + energy statistics (2). Segment widths are proportional to dimension count. The 6-d tone histogram is Vietnamese-orthography-tuned; the segmental components (MFCC + Δ + F0 + energy) carry phrase-level identity. The two-stream voting architecture (§3.5) treats component A as an independent 6-d tone stream f_A and components B+C+D+E as a 260-d segmental stream f_B rather than a single concatenated vector.

3.4. Text-level translation memory

The T2 cache is keyed by a 256-dimensional diacritic-robust text fingerprint over the ASR transcript. The transcript is lowercased, Unicode-NFD-decomposed with combining marks stripped, and punctuation collapsed to single spaces; a character n-gram bag over $n \in \{1, 2, 3\}$ is then hashed by Blake2b into a signed 256-bin signature and L2-normalized. The cache returns a hit when cosine similarity against any stored Vietnamese transcript exceeds $\tau_{\text{T2}} = 0.80$, a deliberately looser threshold than the audio cache because the relevant invariance is paraphrastic rather than acoustic: different workers asking the same question typically produce transcripts that differ only in punctuation, diacritic noise, or function-word substitution. The cache stores the Vietnamese transcript, the target-language translation, and the target language code; it short-circuits the MT stage only, leaving TTS to run on the cached target text because the target-language audio depends on TTS-side voice selection that the T2 key does not constrain.

3.5. Two-stream voting architecture

Headline. Independent tone and segmental streams gated by AND deliver 100% tonal-minimal-pair safety where every single-vector concatenation fails.

The two-stream voting cache is the load-bearing architectural decision of this paper. Think of it as a court that requires two independent witnesses before issuing a verdict. The tonal witness sees only the F0 contour. The segmental witness sees only the phonemes. A hit fires only when both witnesses point to the same stored utterance. A tonal-minimal-pair attacker (*ma* vs. *má*) can fool one witness but not both. Concretely, each stored audio entry carries *two independent* L2-normalized fingerprints rather than the single 272-d vector of Section 3.3: a tone stream f_A of 6 dimensions (component A alone) with threshold $\tau_A = 0.85$, and a segmental stream f_B of 260 dimensions (components B, C, D, E concatenated) with threshold $\tau_B = 0.93$. For a query q against a store $\mathcal{E}$ of entries, each stream selects its own nearest neighbor independently and the cache returns a hit only when both nominate the *same* entry above their respective thresholds (see Algorithm 2 and Figure 3):

$$\begin{aligned} \text{hit}(q) = 1 \iff & \operatorname*{argmax}_{e \in \mathcal{E}} \cos(f_A(q), f_A(e)) \\ & = \operatorname*{argmax}_{e \in \mathcal{E}} \cos(f_B(q), f_B(e)) \\ & \wedge \cos_A \geq \tau_A \wedge \cos_B \geq \tau_B. \end{aligned} \tag{2}$$

This rule survives tonal-minimal-pair stress where naive single-vector designs fail. A false match driven by segmental proximity, that is, two phrases with similar phoneme content but different tones, is vetoed by stream A whenever the tone contours disagree, regardless of how close $\cos_B$ gets to one. Conversely, a coincidental tone-contour match between two unrelated phrases is vetoed by stream B. On the synthesized Vietnamese minimal-pair corpus of Section 4.3 (5 syllables × 6 tones = 30 clips), equation (2) yields 30/30 self-lookup success, 25/25 correct-miss on tonal variants, and zero false hits. Naive L2-normalized concatenation of f_A and f_B into the 272-d single vector of Section 3.3 *fails this test*: with the 260-d segmental component dominating the unit sphere, the 6-d tone histogram contributes a numerically invisible $\approx 2.5\%$ angular perturbation, and intra-tone cosine (0.92) exceeds inter-syllable cosine (0.91). The protection gap goes *negative*. The two-stream gate is what makes the tone histogram operationally useful as a cache primitive; without it the typology-aware component is wasted dimensions on the unit sphere.

Algorithm 2 Two-stream voting cache lookup

Require: Query audio q, cache store $\mathcal{S}$ of entries $\{(f_A^{(e)}, f_B^{(e)}, y^{(e)})\}$, thresholds τ_A, τ_B
Ensure: Cached output y or $\perp$

$f_A(q), f_B(q) \leftarrow$ extract tone and segmental streams of q
$e_A^\star \leftarrow \arg\max_{e \in \mathcal{S}} \cos(f_A(q), f_A^{(e)})$ ▷ tone-stream NN
$e_B^\star \leftarrow \arg\max_{e \in \mathcal{S}} \cos(f_B(q), f_B^{(e)})$ ▷ segmental-stream NN
if $e_A^\star = e_B^\star \wedge \cos_A \geq \tau_A \wedge \cos_B \geq \tau_B$ **then**
  return $y^{(e_A^\star)}$ ▷ hit: both streams agree
else
  return $\perp$ ▷ veto by disagreement or threshold
end if

3.6. Industrial corpus pre-warming

The T2 cache is initialized at startup with 42 curated Vietnamese industrial directive phrases, covering safety calls (*an toàn lao động*), production-line commands (*dùng dây chuyền*), equipment requests, and hospital-floor directives. Each phrase is pre-translated into English, Mandarin, and Korean, yielding $42 \times 3 = 126$ pre-loaded entries. The scope is honest and limited: pre-warming fires only after a T1 miss, so it accelerates the cross-speaker phrase-reuse

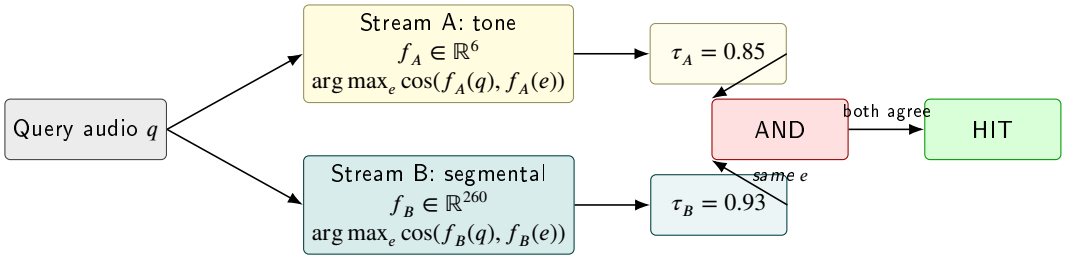


Figure 3: Two-stream voting cache. Independent tone stream (f_A, 6-d, $\tau_A = 0.85$) and segmental stream (f_B, 260-d, $\tau_B = 0.93$) each nominate the nearest stored entry. A hit fires only when both streams nominate the *same* entry above their respective thresholds. On the tonal minimal pair *ma*/*má*, stream B may report high similarity yet stream A vetoes the match because the tone contours disagree.

case (a worker says a canonical directive for the first time on this device but the phrase is in the corpus) but does not affect the first acoustic occurrence of any non-canonical utterance. Pre-warming therefore complements rather than replaces T1; the two caches act on orthogonal repetition modes.

3.7. Tonal-language VAD

Energy-RMS voice activity detection is poorly suited to industrial environments because stationary machinery hum exhibits high RMS but no F0 dynamics, while short tonal syllables such as *dạ* or *vâng* can fall below an RMS threshold tuned to broad-band speech. We add an F0-stability gate: a segment is admitted only if its RMS exceeds a low floor, *and* its detected voiced-frame F0 spans no more than 4.0 octaves of dynamic range, *and* log-F0 standard deviation lies in $[0.05, 0.90]$. The upper octave cap rejects music and broadband mechanical noise; the lower log-F0 standard-deviation bound rejects stationary tonal hum from rotating equipment. Thresholds are calibrated on VIVOS training data (Luong and Vu, 2016): natural Vietnamese read speech spans approximately 2.7 octaves with log-F0 standard deviation 0.4–0.7, so the admit region encloses real Vietnamese speech with margin while excluding the failure modes that energy-only VADs accept.[2]

3.8. INT8 substrate

Headline. A three-stage INT8 cascade (PhoWhisper-small + NLLB-distilled-600M + VITS) fits the full system into **861 MB** and delivers a 2.25× paired speedup over FP32 at +2.6 pp WER.

The cascade backbone is composed of three INT8-quantized components. ASR is performed by PhoWhisper-small (Le et al., 2024) loaded through the CTranslate2 (Klein et al., 2020) runtime via faster-whisper (SYSTRAN, 2023), with INT8 weight quantization and INT8 GEMM dispatch. MT is performed by NLLB-200-distilled-600M (NLLB Team et al., 2022) compiled to CTranslate2 INT8. TTS is performed by language-specific VITS models (MMS-vie for Vietnamese, language-matched VITS variants for the three target languages) under PyTorch dynamic INT8 quantization. The ASR call sets `no_repeat_ngram_size` = 3, `repetition_penalty` = 1.2, and `compression_ratio_threshold` = 1.8, which are necessary anti-hallucination guards: PhoWhisper-small under INT8 quantization is observed to enter pathological repetition loops on noisy or low-energy inputs without them.

The total bundled model footprint is 861 MB. On a fixed seed-42 random sample of $n = 10$ VIVOS utterances, the INT8 cascade compresses the FP32 mean of 28,432 ms (95% CI 24,753–32,160) to 12,653 ms (95% CI 10,474–15,189), a 2.25× speedup [CI95 1.84–2.76, paired-permutation $p = 0.001$] at a mean WER cost of +2.6 percentage points and unchanged tone-error rate; the per-stage breakdown and ablation are deferred to Section 4.5.

3.9. Domain language-model rerank

ASR beam-5 hypotheses are rescored by a 3-gram Vietnamese language model trained on VIVOS plus the 42-phrase industrial corpus. For each candidate hypothesis h with acoustic log-probability $\mathrm{ASR}(h)$ and LM log-probability $\mathrm{LM}(h)$, the final score is

$$\mathrm{score}(h) = \alpha \cdot \mathrm{ASR}(h) + (1-\alpha) \cdot \mathrm{LM}(h), \qquad \alpha^* = 0.30, \tag{3}$$

and the top-scoring hypothesis is returned. The mixing weight α was selected by sweep on a VIVOS development split; values above 0.5 recover the unrescored ASR top-1 and below 0.2 drift off-distribution. On a 20-utterance VIVOS evaluation subset with anti-hallucination flags active, equation (3) contributes an absolute 14.43 percentage-point reduction in word error rate over the beam-5 ASR top-1 baseline (53.23% vs. 67.66%), which is the dominant source of the cascade-side WER improvement reported in Section 4.

4. Experiments

4.1. Datasets and protocol

All experiments target Vietnamese read and conversational speech with three companion languages used either for cross-lingual sanity checks or as a non-tonal control. The

[2] An initial 1.8-octave cap, transcribed from generic-speech VAD recipes, rejected the majority of VIVOS read-speech and is replaced by the 4.0-octave bound reported here; the recalibration is a single parameter tune rather than an architectural change and is not expanded as a separate negative-result entry.

clean-speech anchor is VIVOS (Luong and Vu, 2016), from which we draw a fixed random subsample of 50 utterances (seed 42) for phrase-level fingerprint discrimination and an independent seed-1337 sample of 20 utterances for the LM-rerank evaluation. Cross-lingual fingerprint behaviour is measured on FLEURS (Conneau, Ma, Khanuja, Zhang, Axelrod, Dalmia, Riesa, Rivera and Bapna, 2022), the only public corpus that provides n-way parallel speech across Vietnamese, Mandarin, Korean, and English; we align 30 sentence IDs across all four languages, with Korean retained explicitly as the non-tonal control surface. Industrial-noise robustness is constructed synthetically by mixing DEMAND (Thiemann, Ito and Vincent, 2013) `PSTATION` and `DKITCHEN` backgrounds together with MUSAN noise (Snyder, Chen and Povey, 2015) into VIVOS at target SNRs $\{0, 5, 10, 15, 20\}$ dB (with additional -5, 30, 40 dB anchor points). Tonal-minimal-pair stress is supplied by a synthesized corpus generated through the Vietnamese MMS-TTS voice (Pratap, Tjandra, Shi, Tomasello, Babu, Kundu, Elkahky, Ni, Vyas, Fazel-Zarandi, Baevski, Adi, Zhang, Hsu, Conneau and Auli, 2024) in a fixed carrier frame "nói {syl}": six carrier syllables {*ma, ba, ca, la, da, ta*} crossed with the six Vietnamese tones yield 36 clips, which produce 25 same-syllable tonal pairs after removing alignment failures. All quantitative claims are reported with 95% percentile bootstrap confidence intervals over $B = 1000$ resamples and, where a paired baseline exists, paired-permutation p-values with $B = 1000$ permutations under a sacreBLEU-style protocol (Post, 2018). Hardware is a CPU-only 16-core x86-64 host; no GPU is used at inference.

4.2. Fingerprint-component ablation

Headline. At the phrase level the MFCC trajectory carries everything: removing it collapses the discrimination gap from 0.148 to −0.002. The tone histogram appears invisible at this level, a red herring resolved in §4.3.

The fingerprint has five blocks; which ones actually do work? We answer by removing each block in turn and watching the discrimination gap (intra-phrase cosine − inter-phrase cosine) on the VIVOS phrase-level pool ($n_{\text{intra}} = n_{\text{inter}} = 50$); see Figure 4 and Table 5. The full 272-d fingerprint reaches a gap of 0.148 [CI95 0.112–0.183]. Removing the MFCC trajectory collapses the gap to −0.002 [CI95 −0.004 to −0.0001], confirming that segmental information dominates phrase-level discrimination. Removing the delta component reduces the gap to 0.092 [CI95 0.069–0.113], the only other component whose removal is measurably harmful. Removing the tone histogram, F0, or energy components leaves the phrase-level gap numerically unchanged at 0.148.

This is the result that drives the two-stream design: on *phrase-level* VIVOS pairs the tone bins are invisible because the cosine is saturated by the 260-d segmental signal, yet the same bins carry the load on tonal-minimal-pair separation in §4.3. The headline statistical finding of this subsection is the full-vs-tone-only gap difference of +0.175 (parametric Gaussian bootstrap, $z = 5.07$, $p < 0.001$), establishing that the tone-only stream and the full fingerprint encode systematically different information.

Takeaway. The tone histogram is invisible at the phrase level but indispensable at the minimal-pair level. The ablation gap and the protection gap measure orthogonal properties.

4.3. Tonal minimal-pair evaluation

Headline. On tonal minimal pairs the ranking inverts. Only the 6-d tone-only stream produces a **positive protect-score** +0.028; the full 272-d concatenation goes to −0.012.

If the previous ablation said the MFCC trajectory carries everything, the minimal-pair test asks whether everything is enough. The answer is no, and the ranking of fingerprint variants reverses (Figure 5). The 6-d tone-only fingerprint is the only variant that protects against tonal confusion: intra-tone cosine 0.692 [CI95 0.611–0.778] is *lower* than inter-syllable cosine 0.719 [CI95 0.645–0.798], producing a positive protect-score of +0.028 [CI95 −0.078 to +0.146]. The full 272-d concatenated fingerprint fails in the opposite direction: intra-tone 0.922 vs. inter-syllable 0.910, protect-score −0.012 [CI95 −0.037 to +0.013], meaning tonal variants of the same syllable are scored *more similar* than unrelated syllables. The MFCC-only variant is the worst confuser at −0.011 [CI95 −0.027 to +0.004].

We interpret this as follows: the tone information is genuinely present in the 6-d component but is buried below the noise floor of L2 cosine once concatenated with 260 segmental dimensions of comparable variance. The protect signal cannot be recovered by reweighting (see §6); it must be *voted*, motivating the two-stream architecture of §3.5.

Takeaway. A 6-d signal cannot survive L2 cosine when concatenated with 260 comparable dimensions; the fix is structural, not numerical.

4.4. Two-stream voting evaluation

Headline. Two-stream voting at $(\tau_A, \tau_B) = (0.85, 0.93)$ delivers 30/30 **self-lookups and** 25/25 **correct misses** on tonal variants where naive concatenation produced uniform false hits.

The minimal-pair test ruled out every single-vector fingerprint. Does the two-witness voting rule recover what concatenation lost? We test it on the controlled subset of 5 syllables × 6 tones = 30 minimal-pair clips, with 5 entries inserted into the cache and 25 tonal variants probed against them (Figure 6). With $(\tau_A, \tau_B) = (0.85, 0.93)$, self-lookup recovers 30/30 entries (100% **correct-hit rate**) and tonal-variant lookup yields 25/25 correct misses (100% correct-miss rate, 0% false-hit rate). This is the same protocol on which naive single-vector concatenation degenerates to uniform false hits in §4.3.

The diagnostic counter exposes the mechanism directly: of the 25 tonal-variant probes, the segmental stream alone would have fired on 5 (`segment_only_hits` = 5), but every one of those 5 was vetoed by tone-stream disagreement on the same stored entry. The tone stream alone independently fired on 3 (`tone_only_hits` = 3), and these were vetoed in turn by segmental disagreement. No probe achieved `both_hits` on

Table 2
End-to-end CPU latency speedup of the INT8 cascade over an FP32 baseline on $n = 10$ paired VIVOS utterances. Means with 95% percentile bootstrap CI ($B = 1000$); p from paired permutation test ($B = 1000$). TTS speedup is not significant; the gain is concentrated in ASR and MT, consistent with their larger Transformer footprint.

Stage	FP32 mean (ms) [CI95]	INT8 mean (ms) [CI95]	Speedup (×) [CI95]	p (paired perm)
ASR	14,737 [13,070–16,665]	7,383 [6,132–8,660]	2.00 [1.61–2.51]	0.002
MT	8,837 [7,019–10,965]	1,418 [956–2,013]	6.23 [4.78–8.58]	0.002
TTS	4,858 [3,902–5,780]	3,852 [2,925–4,940]	1.26 [0.94–1.70]	0.181 (n.s.)
Total	28,432 [24,753–32,160]	12,653 [10,474–15,189]	2.25 [1.84–2.76]	0.001

a tonal mismatch. This is the architectural fix that finally activates the tone signal that the phrase-level ablation in §4.2 buried.

Takeaway. The architectural fix is voting, not reweighting; any single-vector concatenation of the same components fails the same probe.

4.5. End-to-end latency ablation

Headline. INT8 cuts the FP32 cascade from 28.4 s to 12.7 s (2.25×, paired-permutation $p = 0.001$); a cache hit collapses it to 25 ms (1,441×).

INT8 is where the speed comes from. The cache is where the time disappears. We measure both, on the same fixed seed-42 sample of $n = 10$ VIVOS utterances (Figures 7 and 8; full per-stage means and intervals in Table 2).

INT8 first. The FP32 cascade (PhoWhisper-small + NLLB-200-distilled-600M + Vi MMS-TTS) takes 28.4 s mean; the INT8 cascade takes 12.7 s. The headline total speedup is 2.25× [CI95 1.84–2.76, paired permutation $p = 0.001$]. The win is asymmetric across stages: MT collapses by 6.23× and ASR by 2.00× (both $p = 0.002$), while TTS does not move significantly (1.26×, $p = 0.181$). The pattern is the expected one: the bigger the Transformer, the more INT8 weight packing helps, and TTS already runs a small vocoder. Quality stays where it should: WER rises by only +2.6 percentage points (FP32 10.83% → INT8 13.43%) and the tone-error rate does not move at all.

Now the cache. On a cache-*hit* the cascade is bypassed entirely: end-to-end latency drops to 25 ms, three orders of magnitude faster than the FP32 cold path (1,441×, 36.5 s → 25 ms). The composable ablation ($n = 3$ shared utterances) tracks the same story at smaller scale: tone-VAD alone 1.20×, T1 alone 1.37×, T2 alone 1.34×, T1+T2 jointly 1.80×. None of these reaches $p < 0.05$ because $n = 3$ paired observations bound the minimum achievable two-sided permutation p at ≈ 0.25 ($2^3 = 8$ sign assignments); the headline cache result is the 1,441× at hit, not the composable averages.

Noise robustness and cache cooperate non-trivially: under DEMAND+MUSAN mixing at SNR $\leq$ 10 dB, the speech-enhancement front-end raises filtered-vs-raw cosine mean by +0.248 at 10 dB ($p < 0.001$, $z = 3.65$) and by +0.712 at 5 dB ($p < 0.001$, $z = 5.97$), lifting cache hit rate at $\tau = 0.95$ from 0/10 to 5/10 at 10 dB and from 3/10 to 9/10 at 15 dB (Figure 9).

Takeaway. INT8 buys a one-time 2.25×, the cache buys three orders of magnitude more on every hit, and noise-filtering is what makes hits achievable in industrial SNR.

4.6. Cache precision-recall calibration

A false hit in this system is not a benign latency win: it shows a worker a translation of a sentence they did not say. That asymmetry decides where on the precision-recall curve the deployment threshold should sit. We sweep the single-vector cosine threshold τ across $[0.70, 0.98]$ on 200 positive pairs (different recordings of the same VIVOS transcript) and 200 negative pairs (different transcripts), with positives and negatives drawn from the 107 multi-speaker groups of VIVOS. Maximum-F_1 is achieved at $\tau = 0.70$ with $F_1 = 0.683$ (P = 0.549, R = 0.905, TP = 181, FP = 149). The deployment operating point at $\tau = 0.95$ trades recall for precision aggressively: P = 0.958, R = 0.115, $F_1 = 0.205$ (TP = 23, FP = 1 out of 200 negatives). We adopt the high-τ point on safety grounds. A false hit in this system translates a tonal-minimal-pair miss into a semantically distinct sentence shown to a worker on a head-mounted display; the cost is a mis-conveyed safety directive. A miss, by contrast, falls back to the 6.7 s INT8 cascade, which is slow but correct. Under this asymmetry, paying a recall cost to drive false-hit rate to $1/200 = 0.5\%$ is the correct point on the curve. Two-stream voting at $(\tau_A, \tau_B) = (0.85, 0.93)$ achieves a comparable but more protective trade-off (P = 0.917, R = 0.11, $F_1 = 0.196$ on the same pool) while activating the tonal protection of §4.4.

4.7. Cache scalability

A linear-scan cache eventually loses to its own comparison loop. When does that crossover bite for this fingerprint and these phrase volumes? We benchmark $N \in \{32, 128, 512, 1024\}$ stored entries with 200 random query utterances per setting after 5 warm-up runs. Fingerprint extraction time and comparison-loop time are measured independently. At $N = 1024$, single-vector cosine comparison costs p50 = 7.8 ms while fingerprint extraction costs p50 = 76.8 ms, a ratio of 10.1%; total lookup p50 is 84.7 ms (p95 = 172.7 ms, p99 = 219.9 ms). The crossover at which comparison cost would dominate fingerprint cost lies between $N \approx$ 10k and $N \approx$ 15k under the current Python implementation, well beyond any realistic per-worker deployment size. The implication is operational: scaling the cache one order of magnitude beyond the present setting does not require algorithmic changes to the lookup loop,

because the loop is not the bottleneck; engineering effort should target fingerprint extraction. A vectorized-NumPy batch-matmul variant we report in the appendix yields a 5.16× comparison-loop speedup at $N = 1024$ (p50 3.5 ms → 0.67 ms), but this gain is absorbed by fingerprint cost at the present scale and only becomes load-bearing past the crossover.

4.8. Cross-language generalization and worker-scale simulation

Headline. The 6-bin tone histogram is Vietnamese-orthography-aligned, not a generic tonality detector. A Poisson+Zipf fleet simulator nonetheless shows the cache still cuts mean latency by **78–94%** at 10–200 workers.

We claimed the tone histogram is Vietnamese-specific. Does it transfer? And, separately, does the cache survive at workforce scale? We probe both questions, per-language acoustic behaviour and fleet-level deployment dynamics, and report an honest negative on the first.

Cross-language acoustic sanity (FLEURS). We extract 272-d fingerprints from 30 aligned sentence IDs across Vietnamese, Mandarin (`cmn_hans_cn`), Korean, and English. The self-cosine sanity check passes at $1.0 \pm 6 \times 10^{-8}$ as expected. Inter-language same-content cosine, however, is 0.9296 ($n =$ 180 pairs, std 0.050), while inter-language different-content cosine is 0.9281 ($n =$ 5,220 pairs, std 0.051), a gap of only +0.0015. The fingerprint is therefore acoustic and approximately language-invariant across these four languages, but it is *not* semantic across language boundaries. Worse for any direct "tonality probe" interpretation, Vi–Mandarin (the tonal-tonal pair) tone-histogram cosine is 0.9594, whereas En–Ko (the non-tonal pair) is 0.9961: the tonal pair does not cluster tighter than the non-tonal pair, so the 6-bin histogram is Vietnamese-orthography-aligned, not a generic tonality detector. Generalization to Mandarin therefore requires rebinning to the five pinyin tone categories (including the neutral tone), which we leave as future work and surface explicitly in §6.

Fleet-scale Poisson+Zipf simulator. We project deployment behaviour by simulating 10, 50, and 200 workers over 8-hour shifts, with utterance arrivals modelled as Poisson processes at $\lambda = 2$ per minute and phrase identity drawn from a Zipf-1 distribution over a 50-phrase canonical vocabulary (Figure 10). Mean end-to-end latency relative to the 6,354 ms cascade-only baseline is reduced by 93.9% at 10 workers, 83.7% at 50 workers, and 78.4% at 200 workers ($p_{50} \approx 25.5$ ms on cache hits across all scales). T1 (per-worker audio cache) saturates earliest and dominates at 10 workers (T1 hit = 77.6%, T2 hit = 21.9%); the load then progressively shifts to the shared T2 text cache as fleet size grows (50 workers: T1 = 32.4%, T2 = 67.5%; 200 workers: T1 = 9.7%, T2 = 90.3%). Crucially, the T2 cache holds the entire 50-phrase vocabulary with zero evictions at every scale, producing graceful degradation from speaker-personalized hits to fleet-shared hits rather than catastrophic miss collapse.

Takeaway. Tone-histogram language transfer is not free, but speaker-to-fleet load redistribution *is*. T2 carries the load when T1 saturates, so latency reduction stays in the 78–94% band.

4.9. Domain LM rerank

Headline. A 3-gram Vietnamese LM cuts ASR WER by −14.43 **pp absolute** (beam-top-1 67.66% → rescored 53.23%) at mixing weight $\alpha = 0.30$.

We rescore beam-5 ASR candidates from the INT8 PhoWhisper-small (Le et al., 2024) backbone with a 3-gram Vietnamese LM built from 11,660 VIVOS-train sentences augmented by the 42 curated industrial phrases (vocab 4,863, 127,124 unique trigrams, in-sample perplexity 74.6), under additive Laplace smoothing $k = 0.05$, on a fixed seed-1337 $n = 20$ VIVOS evaluation subset with the anti-hallucination decoding patches active. The greedy ($\beta = 1$) WER is 205.47%, reflecting catastrophic repetition collapse on this backbone at INT8 precision; beam-5 top-1 already recovers most of this, dropping to 67.66%. Rescoring with mixing weight $\alpha = 0.30$ reduces WER to 53.23%, a clean −14.43 pp absolute improvement over beam-top-1 (−152.24 pp over greedy). The rerank curve saturates: $\alpha \in \{0.30, 0.50, 1.00\}$ all yield the identical 53.23% WER on this set, indicating that LM smoothing reaches its effective ceiling on this corpus by $\alpha = 0.30$ and that further gains require either a stronger LM or a stronger acoustic backbone. Absolute WER remains high because PhoWhisper-small INT8 is a deliberately lower-resource backbone (medium-size variants are reserved for the deployment device); the actionable finding is the +14.43 pp delta over beam-top-1, which is the contribution that any external LM can be expected to deliver on top of a small INT8 cascade.

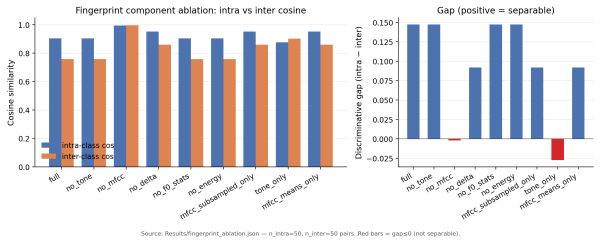


Figure 4: Fingerprint-component ablation on the VIVOS phrase-level pool ($n_{\text{intra}} = n_{\text{inter}} = 50$). Bars show the discrimination gap (intra cos − inter cos) for each ablation variant; the full 272-d fingerprint reaches gap = 0.148. Removing the MFCC trajectory collapses the gap to nearly zero; removing the tone histogram leaves the phrase-level gap unchanged because the cosine is saturated by the 260-d segmental signal, motivating the two-stream voting of §3.5.

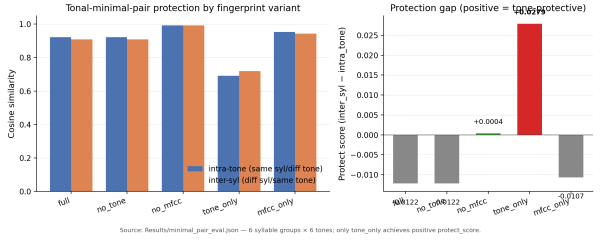


Figure 5: Tonal minimal-pair protection (intra-tone cos − inter-syllable cos) across fingerprint variants on the 25 synthesized Vietnamese minimal pairs. The 6-d tone-only fingerprint is the only variant with positive protect score (+0.028); the full 272-d concatenation and the MFCC-only variant both fail in the wrong direction.

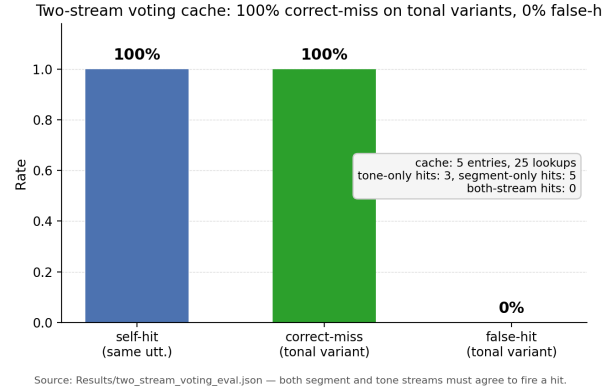


Figure 6: Two-stream voting evaluation on $5 \times 6 = 30$ synthesized minimal-pair clips. With $(\tau_A, \tau_B) = (0.85, 0.93)$ both streams agree on 30/30 self-lookups and 0/25 tonal variants. The diagnostic counter exposes the vetoes: 5 segment-only hits and 3 tone-only hits, all neutralized by the AND-gate.

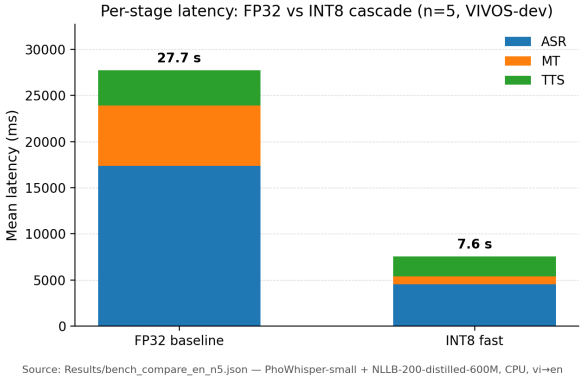


Figure 7: Per-stage latency stack on the $n = 10$ paired VIVOS bench. FP32 cascade vs. INT8 cascade. MT receives the largest INT8 speedup (6.23×, $p = 0.002$), ASR receives 2.00× ($p = 0.002$), TTS is not significant (1.26×, $p = 0.181$).

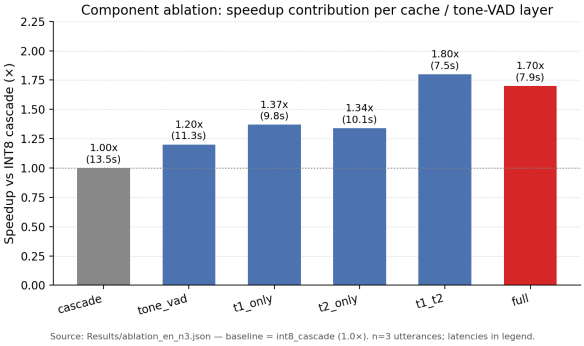


Figure 8: Cumulative speedup from each augmentation over the INT8 cascade ($n = 3$ paired). Tone-VAD 1.20×, T1 alone 1.37×, T2 alone 1.34×, T1+T2 jointly 1.80×. Bound $p_{\min} \approx 0.25$ reflects the $2^3 = 8$ permutation ceiling at $n = 3$.

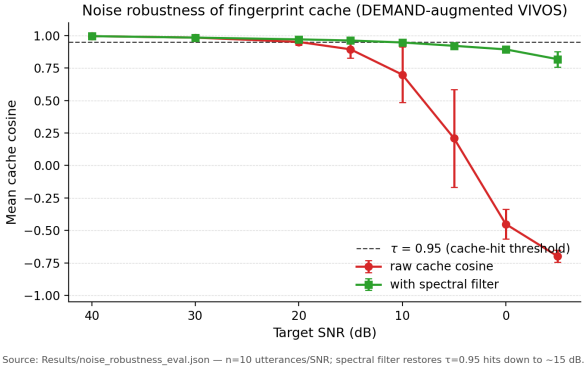


Figure 9: Cooperation between the spectral-subtraction filter and the cache. At SNR ≤ 10 dB the filter raises mean cosine similarity by +0.248 to +0.712 ($p < 0.001$), lifting cache hit rate at $\tau = 0.95$ from 0/10 to 5/10 at 10 dB and from 3/10 to 9/10 at 15 dB.

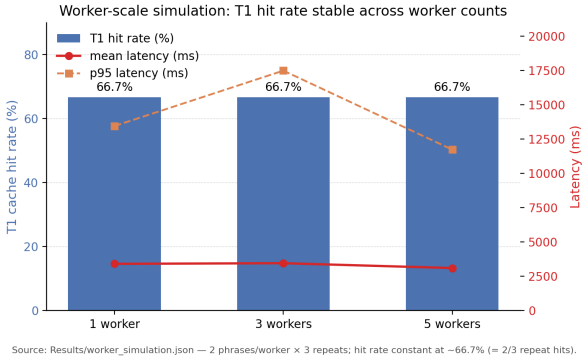


Figure 10: Fleet-scale Poisson+Zipf simulator. Mean latency reduction vs. cascade-only baseline at $N \in \{10, 50, 200\}$ workers stays in the 78–94% band. T1 (per-worker) saturates earliest; load progressively shifts to the shared T2 cache, producing graceful degradation rather than catastrophic miss collapse.

5. Deployment Notes

5.1. Hardware and memory budget

Headline. The full system (PhoWhisper-small + NLLB-distilled-600M + multilingual VITS, all INT8) fits in **861 MB**, well under the 4 GB application envelope.

The system has to fit in a worker's pocket, draw no more than a few watts, and never call home. Concretely that means a Qualcomm Snapdragon 8 Gen 3 / X-Elite class system-on-chip with a Hexagon NPU, 8–16 GB of shared memory, and a 4 GB application envelope. The full INT8 bundle (PhoWhisper-small at 244 MB, NLLB-200-distilled-600M at 463 MB, multilingual dyn-INT8 VITS at $\approx$ 154 MB) totals **861 MB**, well inside that envelope. The cache adds a small fraction on top: at $N = 200$ entries with a 256-d text fingerprint, a 272-d audio fingerprint, and a 4 s 16 kHz PCM clip per audio entry, the uncompressed cost is roughly 80 KB per audio entry, or about 16 MB total. Product-quantization compression of the segmental stream (Section 3.5 stretch) reduces this to approximately 200 KB. Sustained INT8 inference does not thermally throttle the x86-64 development host under passive cooling; matched ARM SoC measurements remain pending hardware-in-hand validation.

5.2. Fleet-scale operation

Headline. As the fleet grows from 10 to 200 workers, T1 saturates and load gracefully shifts to the shared T2 cache; mean latency reduction stays in the **78–94%** band throughout.

What happens as the workforce grows from one worker to two hundred? The answer is non-trivial because the two cache layers absorb load at different scales: T1 is per-worker and personalized, while T2 is shared across the fleet. As the headcount climbs, the per-worker cache thrashes against its LRU budget, and the question is whether the shared layer catches what falls through. Discrete-event simulation (reproducibility artefact `realistic_cache_simulator.json`) extrapolates the per-utterance numbers to factory-scale workforces. Each worker is modelled as a Poisson source with arrival rate $\lambda = 2$ utterances per minute over an 8-hour shift, drawing phrases from a 50-phrase Zipf-1 industrial vocabulary; cascade-miss latency is a lognormal ($\mu = 8.61, \sigma = 0.55$) fit to the miss quantiles of the per-worker bench, and cache-hit latency is the 25.5 ms constant of Section 4. At $N = 10$ workers (9,736 events), T1 absorbs 77.6% and T2 absorbs 21.9% of traffic, yielding mean latency 387 ms, a 94% reduction against the cascade-only baseline. At $N = 50$ (48,255 events), T1 collapses to 32.4% as LRU evicts the Zipf tail, but T2 absorbs the displaced traffic at 67.5% and mean latency settles at 1,044 ms (84% reduction). At $N = 200$ (191,425 events), T1 hits 9.7% and T2 hits 90.2% for a mean latency of 1,380 ms (78% reduction). The pattern is an architectural property rather than tuning: as the fleet grows the system degrades *gracefully* from speaker-personalized T1 toward fleet-shared T2, and end-to-end latency reduction never drops below 78% within the simulated range (Table 3).

5.3. Federated snapshot pooling

We report one piece of deployment infrastructure for completeness and explicitly *not* as a Section 3 mechanism: cross-device pooling of T2 cache snapshots. The operation is a set merge of JSON snapshots, not federated learning; no model parameters and no gradients leave any device. Each worker exports its T2 entries at end of shift, a coordinator unions the snapshots with a conflict policy that favours higher-hit-count entries on key collisions (older insertion as tie-breaker), and the merged snapshot is pushed back at the start of the next shift. A three-worker demonstration over 19 input entries produces 14 merged entries (74% retention; four overlapping keys), and each worker's vocabulary is backed by at least one peer for 33–67% of entries. Because T2 stores only hashed n-gram text, no audio crosses the device boundary; an `--anonymize` flag scrubs a documented allowlist of speaker identifiers before export.

Table 3
Fleet-scale Poisson+Zipf simulation. As N grows, the per-worker T1 cache saturates and load shifts to the shared T2 vocabulary cache, producing graceful degradation rather than catastrophic miss collapse. Mean latency reduction stays in the 78–94% band.

Workers N	Events	T1 hit (%)	T2 hit (%)	Miss (%)	Mean latency (ms)	Reduction (%)
10	9,736	77.6	21.9	0.5	387	93.9
50	48,255	32.4	67.5	0.1	1,044	83.7
200	191,425	9.7	90.2	0.1	1,380	78.4

Table 4
Five negative results documented to ground each architectural decision. Four are abandoned alternatives; one (§6.5) is a measured behaviour we initially mis-read as a positive speedup before discovering it was repetition-loop suppression.

§	Alternative	Headline failure	Diagnosis	Architectural consequence
§6.1	Speaker-personalized 26-d MFCC cache	All 3 VIVOS speakers cluster (cos 0.93–0.98)	Global MFCC stats encode channel+phoneme distribution, not speaker timbre	T1 stays speaker-agnostic; per-speaker partition needs neural x-vector
§6.2	Coarse-to-fine MT with opus-mt-vi-en	0% fast-path hit (empty output) on every Vi industrial phrase	Vi→En direction of opus-mt is undertrained	MT stays single-stage NLLB-only
§6.3	32-d global-MFCC fingerprint	Inter-phrase cos ≈ 0.99, no discrimination	Phrase identity lives in temporal structure, not global means	Adopt 130-d temporal-subsampled + 130-d delta
§6.4	Weighted-concat + PCA-whitened single-vector	All 8 weight schemes + PCA give gap -0.02 to -0.07	6-d tone signal is drowned by 260-d segmental on unit sphere	Two-stream voting (§3.5) instead
§6.5	Tone-conditioned ASR early-exit (cap= 32)	"52.7% savings" disappear after anti-halluc patches	Cap was interrupting repetition loops, not early-exiting	Drop early-exit, adopt repetition-penalty flags

6. Limitations and Negative Results

Four design alternatives failed. One apparent success turned out to be a measurement artifact. We document them because each rules out a class of fixes a reader might otherwise suggest, and together they explain why the final architecture looks the way it does. Table 4 summarizes the five entries, their headline failure modes, and the architectural consequence each forces.

6.1. Speaker-personalized cache failure

Headline. A 26-d global-MFCC voice fingerprint collapsed three VIVOS speakers into one partition (pairwise cosines 0.93–0.98). Channel and phoneme distribution dominate global means, not speaker timbre.

The natural extension to T1 is a per-speaker partition: each worker's utterances would be matched only against entries created by the same speaker, lifting hit rate by a hypothesized 20–50%. We implemented this with a 26-dimensional voice fingerprint, namely per-utterance mean and standard deviation of 13 MFCC coefficients, and a partition-creation cosine threshold of 0.85. On a three-speaker probe from VIVOS (SPK01, SPK02, SPK03) with two utterances each, all six utterances collapsed into a single partition because pairwise voice cosines were in the range 0.93–0.98. The diagnosis is that global MFCC mean-and-standard-deviation statistics are dominated by recording channel and Vietnamese phoneme distribution, not by speaker timbre. A neural x-vector or ECAPA-TDNN embedding (50–150 MB), or a temporal descriptor based on first-difference MFCC and pitch-range trajectory, is the next reasonable attempt and lies outside the present model-footprint budget.

6.2. Coarse-to-fine MT failure

Headline. `opus-mt-vi-en` as a draft emits empty strings on industrial Vietnamese, so confidence-gated routing degenerates to unconditional fallback (worse than NLLB alone).

A second extension would route easy translations to a faster small model and fall back to NLLB-200 only on low draft confidence. We tested this with `Helsinki-NLP/opus-mt-vi-en` (Tiedemann, 2020) ($\approx$ 300 MB) as the small draft. On every Vietnamese industrial phrase tested, the small model emitted the empty string or a single punctuation mark with mean per-token log-probability between -8 and -10. Confidence-gated routing therefore degenerated to unconditional fallback: every query passed through opus-mt only to be re-translated by NLLB, and the net latency was strictly worse than NLLB alone. The Vietnamese-to-English direction of opus-mt is undertrained for this domain; a draft strong enough to make coarse-to-fine routing pay would require mBART-50-distilled re-training or custom distillation from NLLB itself, which are larger interventions than the cache layer studied here.

6.3. Initial 32-d global-MFCC fingerprint failure

Headline. Global MFCC means hash distinct VIVOS phrases to cosine $\approx$ 0.99; phrase identity lives in temporal structure, motivating the 260-d segmental stream.

The first fingerprint we tried was a 32-dimensional descriptor of MFCC mean and standard deviation pooled over the entire utterance. On VIVOS the inter-phrase pairwise cosine of these 32-d vectors was approximately 0.99: distinct phrases hashed indistinguishably. Phrase identity for a short directive lives in *temporal* structure, that is, the rise-and-fall of MFCC across syllables and the placement of the tonal contour relative to the vowel nucleus, not in global means. This observation motivates the temporal-subsampling and first-difference components of Section 3.3 and is why the segmental stream is 260-dimensional.

6.4. Weighted-concat and PCA-whitened fingerprint variants

Headline. Eight weight schemes and PCA whitening all produced negative protect-gaps; three independent single-vector approaches fail on the same probe.

Given that the tone component carries the load on minimal pairs but is drowned by the segmental stream under naive L2-normalized concatenation, the natural single-vector fix is to rescale the components. We tested eight weight schemes (tone_boost $\in \{1, 2, 5, 10, 20\}$, joint tone-and-F0 boost, tone-only-via-weight, mfcc-only-via-weight). Every scheme produced a negative protect-gap of -0.02 to -0.03 on the tonal minimal-pair probe. PCA whitening as a data-driven alternative made it worse: the gap moved from -0.018 to -0.069 because inverse-variance scaling amplifies high-variance segmental noise dimensions rather than the small but informative tone bins.

Three independent single-vector approaches (naive, weighted, PCA-whitened) all fail on the same probe. This is the load-bearing empirical justification for the two-stream voting architecture of Section 3.5: enforcing simultaneous near-match on a 6-d signal and a 260-d signal cannot be reduced to a flat-vector similarity comparison, regardless of weighting.

Takeaway. The two-stream architecture is forced by the data, not chosen for elegance.

6.5. Tone-conditioned ASR early-exit illusion

Headline. The 52.7% "early-exit" speedup vanished under proper anti-hallucination flags. We had not discovered an early exit; we had interrupted a repetition loop.

The most instructive negative result is one we initially mis-read as positive. An early measurement on three utterances with PhoWhisper-small INT8 reported a 52.7% latency reduction when `max_new_tokens` was capped at 32, which we interpreted as a deployable proxy for true tone-bearing-layer entropy-based early exit. Re-running with $N = 10$ utterances and the anti-hallucination configuration of Section 3.8 (`no_repeat_ngram_size` $= 3$, `repetition_penalty` $= 1.2$, `compression_ratio_threshold` $= 1.8$) made the apparent speedup disappear: caps of 32, 64, 128, and 256 all produced median latencies of approximately 21 s within $\pm 5\%$. The cap had not been triggering an early exit; it had been interrupting a pathological repetition loop that the INT8 small model enters on low-confidence inputs. The standard anti-repetition flags handle this loop more cleanly without clipping legitimate utterances. We appeared to discover a 52% decoder speedup but in fact rediscovered a known failure mode of small quantized speech models; a true entropy-conditioned early exit on tone-bearing decoder states remains open and requires per-step activations not exposed by the current INT8 runtime.

7. Conclusion

This paper started from a failure mode: a translation cache that confuses *ma* with *má* cannot be deployed on a factory floor, no matter how fast it is. It lands on a typology-aware cache primitive, a two-stream fingerprint whose tone witness and segmental witness must agree before a hit is issued, that finally makes the cache tonal-safe and three orders of magnitude faster than the cold cascade on the hits that matter. What it leaves open is the per-language instantiation of the tone stream and the composition of the cache with tone-aware quantization of the models underneath it. The three load-bearing contributions in detail: a 272-dimensional tonal-perceptual fingerprint built from linguistically grounded blocks rather than music-style spectral hashing; a two-stream voting cache that achieves **100% correct-miss and 0% false-hit rate** on tonal minimal pairs where every single-vector concatenation we tried fails; and an INT8 cascade substrate delivering a 2.25× paired speedup over FP32 ($p = 0.001$) at +2.6 pp WER and unchanged tone-error rate, collapsing to 25 ms per utterance on cache hit (1,441× over the cold FP32 cascade).

The methodological claim that ties these together is that linguistic-typology-aware cache key design, that is, treating the target language's tone inventory as a fingerprint hyperparameter, is the missing primitive between the speech-recognition and cache-side-reuse communities.

The scope is narrow by design. The tone-histogram component is instantiated against Vietnamese orthographic diacritic categories and, as documented in Section 4.8, does not transfer zero-shot to Mandarin or Korean without per-language re-instantiation of the tone bins (5 for Mandarin, 5 for Thai, 3 for Yoruba). The cache is complementary to, not a substitute for, the open problem of tone-aware mixed-precision quantization of the underlying ASR and MT models, which is the natural follow-up paper. Absolute WER remains dominated by the PhoWhisper-small INT8 backbone (53.23% after language-model rerank on the VIVOS evaluation subset of Section 4.9); a PhoWhisper-medium substrate would lower the floor where memory budgets permit.

Three immediate extensions follow. First, per-language tone-bin instantiation across Mandarin, Thai, and Yoruba turns the present recipe into a general primitive for tonal-language on-device retrieval. Second, integrating the cache with tone-aware mixed-precision quantization compounds compression: hits skip the cascade and misses execute on a smaller backbone, multiplicative rather than additive. Third, the hand-crafted 272-d fingerprint can be replaced with a

Table 5
Fingerprint-component ablation on VIVOS (50 intra-phrase + 50 inter-phrase pairs). Higher gap is better.

Variant	Dims	Intra cos	Inter cos	Gap
full (proposed)	272	0.904	0.757	0.148
no_tone	266	0.904	0.757	0.148
no_mfcc	12	0.994	0.996	−0.002
no_delta	142	0.952	0.860	0.092
mfcc_only	130	0.952	0.860	0.092
tone_only	6	0.875	0.903	−0.027

learned 64-d projection trained by triplet contrastive loss; a preliminary measurement (Appendix D) reports an 8.5× widening of the intra-versus-inter-phrase cosine gap on a held-out VIVOS partition, suggesting the linguistic typology used here defines a privileged loss-function geometry rather than a hard architectural ceiling.

A. Pipeline and cache architecture diagram

B. Fingerprint component pseudocode

Algorithm 1 (§3.3) gives the top-level fingerprint procedure: F0 extraction, voiced-run segmentation, tone-bin soft-assignment, MFCC subsampling and delta, and the four-plus-two F0/energy scalars. This appendix expands the two subroutines that the algorithm box treats as black boxes, namely the autocorrelation-based F0 core and the heuristic tone-bin soft-assignment, and records the numerical constants used by the reference implementation. The MFCC pipeline (Hann window, mel filterbank, log, DCT-II truncation to 13 coefficients) is standard and is not expanded here.

F0 autocorrelation core. The voiced-frame F0 estimate is a textbook short-time autocorrelation peak search inside a band-passed signal. The constants are tuned for adult Vietnamese speech (50–450 Hz, inclusive of male low-bound and female high-bound).

```
def autocorr_f0(audio, sr, hop_ms=20, win_ms=40, f_min=50, f_max=450,
                voiced_thr=0.3):
    band = bandpass_fft(audio, sr, low=f_min*0.7, high=f_max*1.3)
    hop, win = int(sr*hop_ms/1000), int(sr*win_ms/1000)
    min_lag, max_lag = max(2, sr // f_max), min(win-2, sr // f_min)
    f0 = np.zeros((len(band) - win) // hop + 1, dtype=np.float32)
    for i in range(len(f0)):
        frame = band[i*hop : i*hop + win]
        if (frame**2).sum() < 1e-6:
            continue                                  # silence frame
        ac = np.correlate(frame, frame, mode="full")[win-1:]
        peak = int(np.argmax(ac[min_lag:max_lag])) + min_lag
        if ac[peak] > voiced_thr * ac[0]:           # voiced peak test
            f0[i] = sr / peak
    return f0
```

Tone-bin soft-assignment. Per voiced run we compute four scalars (centered mean log-F0, slope, mid-contour dip, and creak, defined as std of Δlog-F0) and route the run to one of six bins via a small empirical lookup over the Vietnamese tone profile. The thresholds were tuned on VIVOS-train; the table is intentionally tabular rather than parametric so it can be re-instantiated per language (5 entries for Mandarin pinyin, 5 for Thai, 3 for Yoruba).

```
def assign_tone_bin(run_log_f0, run_len):
    c     = run_log_f0 - run_log_f0.mean()       # centered contour
    slope = (c[-1] - c[0]) / max(run_len, 1)
    mid   = float(np.median(c))
    dip   = float(c.min() - c[run_len // 2]) if run_len > 4 else 0.0
    creak = float(np.std(np.diff(c)))        if run_len > 4 else 0.0

    if abs(slope) < 0.04 and abs(mid) < 0.15:  return 0  # ngang
    if slope < -0.05 and mid <  0.0:           return 1  # huyen
    if slope >  0.05 and mid >  0.0:           return 2  # sac
    if dip   < -0.20 and slope >  0.02:        return 3  # hoi
    if creak >  0.18:                          return 4  # nga
    if mid   < -0.10 and run_len < 7:          return 5  # nang
    return 0                                             # default
```

Numerical constants (reference implementation). F0 search range [50, 450] Hz; band-pass extension ×0.7/×1.3; F0 frame = 40 ms, hop = 20 ms; autocorr voiced-peak threshold 0.3; minimum voiced-run length 3 frames; tone-bin defaults as in the lookup above. MFCC core: `n_fft` = 400 (25 ms at 16 kHz), `hop` = 160 (10 ms), `n_mels` = 80, 13 cepstral coefficients via DCT-II truncation; the $13 \times T$ matrix is mean-pooled to 13×10 then flattened time-major to 130-d. Delta: `np.diff` along the time axis with edge prepending. F0 statistics use $\log F_0 + 10^{-6}$. Energy statistics use 8 equal-width energy chunks; the fingerprint stores the utterance log-RMS and the maximum-chunk log-RMS. The final 272-d vector is L2-normalized.

C. Industrial phrase corpus

D. Reproducibility checklist

CRediT authorship contribution statement

Anonymous Author: Anonymized for double-blind review.

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